VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI – 590 018

A Major Project Phase-II (BCD786) Report

On

"Smart Road Infrastructure: AI-Powered Pothole Detection and Alert"

Submitted in the partial fulfillment of the requirement for the award of degree

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in

CSE (Data Science)

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SAI VIDYA INSTITUTE OF TECHNOLOGY

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RAJANUKUNTE, BENGALURU – 560 064

2025-26

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CERTIFICATE

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ABSTRACT

Urban road infrastructure plays a critical role in ensuring smooth mobility, public safety, and the efficient delivery of essential services. In rapidly growing cities like Bangalore, recurring issues such as potholes and open manholes continue to disrupt traffic flow, increase vehicle repair costs, and pose significant safety risks to commuters. Traditional road assessment practices rely heavily on manual inspections and citizen complaints—methods that are often slow, labour-intensive, and vulnerable to human error or subjective interpretation.

To overcome these challenges, this paper introduces a Smart Road Infrastructure Monitoring System powered by Artificial Intelligence (AI) and Computer Vision. The framework leverages the Ultralytics YOLOv8 deep-learning model to process visual data collected from CCTV feeds, drones, satellite imagery, and vehicle-mounted cameras. This enables real-time detection and classification of road-surface anomalies, particularly potholes and manholes. Once detected, each anomaly is automatically geotagged using GPS and uploaded to a secure cloud server.

A dedicated web-based dashboard provides authorized stakeholders—including the Greater Bengaluru Authority (GBA)—with intuitive tools to monitor hazard frequency, severity levels, and spatial patterns across the city. This visualization capability supports evidence-driven planning and helps authorities prioritize repair activities, allocate maintenance resources more efficiently, and respond to critical issues without delay. The system further includes automated report-generation features, producing structured summaries of detected anomalies and their exact locations, thereby improving documentation accuracy and streamlining decision-making processes.

By integrating AI-powered detection, cloud computing, and geospatial analytics, the proposed system significantly reduces human dependency, accelerates hazard response times, and promotes greater transparency in urban governance. Beyond immediate operational benefits, the platform also supports long-term infrastructure analysis, enabling authorities to identify recurring hotspots and anticipate future maintenance demands.

Looking ahead, the system can be enhanced through the addition of mobile reporting applications, more advanced anomaly classification capabilities, and automated coordination with municipal repair teams—further strengthening the vision of a proactive, data-driven approach to smart urban infrastructure management.

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CHAPTER-1

INTRODUCTION

1.1 Background and Context

Urbanization is reshaping the world at an unprecedented pace, influencing every aspect of human life—from economic development to social interaction—and transforming the way cities evolve, operate, and sustain their growing populations. According to the United Nations, more than 55% of the global population currently resides in urban areas, and this figure is projected to rise to an astonishing 68% by 2050. This demographic shift signifies a massive transformation in how cities function, demanding stronger systems of mobility, communication, and infrastructure to support increased population densities. Urban centers have emerged as engines of innovation, driving technological advancement, economic prosperity, and societal progress by attracting industries, businesses, and millions of people seeking opportunities.

However, this rapid development brings with it a series of complex challenges. Among these, the construction, expansion, and upkeep of essential infrastructure stand out as one of the most critical components requiring urgent and sustained attention. Within the broader spectrum of infrastructure, road networks are especially vital because they serve as the arteries of any urban environment, enabling the continuous flow of vehicles, goods, services, and people. In India, Bangalore serves as a prime example of the realities of modern urbanization. Over the last two decades, it has evolved from a quiet city into a globally recognized technology hub, earning the title “Silicon Valley of India.” This economic boom has resulted in the development of an extensive network of roads and highways designed to support the increasing traffic load. However, while expansion has been rapid, maintenance has not kept pace, leading to significant deterioration in road quality. Today, Bangalore faces a persistent problem: potholes and road damage that compromise commuter safety and overall transportation efficiency.

According to the Bangalore Development Authority (BDA), the city now has over 13,000 kilometers of roads, yet potholes remain a recurring issue that becomes particularly severe during the monsoon season. Heavy rainfall accelerates road wear, causes waterlogging, and turns even minor cracks into major hazards overnight. Unfortunately, the absence of a proactive monitoring system means repairs are often delayed, allowing the problem to escalate

and posing daily risks to thousands of commuters who rely on Bangalore's road network for work, education, business, and essential services.

1.2 Importance of Road Infrastructure

Road infrastructure is one of the most critical components of urban development. It serves as the foundation for economic growth, trade, and daily commuting. Roads are not merely strips of asphalt and concrete—they are the lifelines of cities, enabling the seamless movement of goods, services, and people. A well-connected and well-maintained road network ensures that businesses thrive, supply chains remain uninterrupted, and citizens can access essential services without delays.

Well-maintained roads offer numerous benefits:

- **Smooth Transportation**: Efficient Road networks reduce travel time and improve connectivity between residential, commercial, and industrial zones.
- **Reduced Fuel Consumption**: Good Road conditions minimize unnecessary braking and acceleration, leading to lower fuel usage and reduced emissions.
- **Enhanced Safety**: Properly maintained roads reduce the risk of accidents caused by potholes, cracks, and uneven surfaces.

Conversely, poor road conditions have severe consequences:

- **Increased Vehicle Maintenance Costs**: Damaged roads lead to frequent repairs and higher maintenance expenses for vehicle owners.
- **Higher Accident Rates**: Potholes pose serious hazards, especially for two-wheelers and pedestrians.
- **Economic Losses**: Traffic congestion and delays caused by poor road conditions result in productivity losses and increased transportation costs.

A study by the **Indian Road Congress** estimates that poor road conditions cost the Indian economy **₹20,000 crore annually** in fuel wastage and vehicle damage. In Bangalore, where traffic congestion is already a major concern, the presence of potholes and amplifies the problem, making road maintenance a critical urban challenge. These hazards not only slow down traffic but also increase the likelihood of accidents, particularly during monsoon seasons when waterlogging conceals road defects.

Road infrastructure is therefore more than just a physical asset—it is a strategic enabler of economic development and social well-being. Cities that invest in robust road maintenance

systems experience improved mobility, reduced environmental impact, and enhanced public safety. In the context of Bangalore's rapid urbanization, ensuring the quality and reliability of road infrastructure is essential for sustaining growth and improving the quality of life for millions of residents.

1.3 Challenges in Current Road Maintenance

Maintaining urban road infrastructure is a complex and resource-intensive task, and traditional approaches have significant limitations. Most cities, including Bangalore, rely heavily on **manual inspections** and **citizen complaints** to identify road damage. While these methods have been in use for decades, they are inherently **reactive**, slow, and inefficient.

➢ Manual Inspections

Inspectors physically visit roads, observe conditions, and record findings—a process that is:

- **Time-consuming**: Covering large urban areas requires enormous manpower and time.
- **Labor-intensive**: Inspectors must walk or drive through long stretches of roads.
- **Prone to human error**: Observations can vary based on individual judgment, leading to inconsistencies.

Because of these limitations, manual inspections often fail to detect issues early. By the time damage is reported, it may have worsened, increasing repair costs and posing greater risks to commuters.

➢ Citizen Complaints

The second major method involves citizens reporting road hazards through helplines, apps, or municipal portals. While this approach encourages public participation, it suffers from:

- **Inconsistency**: Not all citizens report issues promptly.
- **Delays**: Complaints often take days or weeks to reach the right department.
- **Bureaucratic hurdles**: Multiple layers of approval and communication slow down repair work.

➢ Reactive Nature of Traditional Methods

Both manual inspections and citizen complaints are **reactive**—they address problems only after they occur. This delay allows hazards like potholes to persist, increasing the likelihood of accidents and traffic congestion.

➢ Impact of Monsoon Rains

In cities like Bangalore, monsoon rains accelerate road deterioration. Water seeps into cracks, eroding the base layer and creating potholes overnight. Without a proactive monitoring system, these hazards remain undetected until they cause accidents or severe traffic disruptions.

➢ Consequences of Inefficiency

The lack of timely maintenance leads to:

- **Recurring hazards**: Roads repaired without proper monitoring often deteriorate again.
- **Economic losses**: Delays in repair increase fuel consumption and vehicle damage.
- **Public safety risks**: Poor road conditions contribute to accidents, especially for two-wheelers.

1.4 Global Context of Smart Infrastructure

Across the globe, cities are embracing the concept of **smart infrastructure** to address the growing challenges of urbanization. As populations rise and urban spaces expand, traditional methods of infrastructure management are proving inadequate. To overcome these limitations, many countries have turned to advanced technologies such as **IoT sensors, AI algorithms, and cloud platforms** to ensure that roads and transportation systems remain safe, efficient, and sustainable.

Smart infrastructure integrates real-time data collection, predictive analytics, and automated decision-making to create cities that are not only connected but also intelligent. These systems monitor road conditions, traffic flow, and structural health, enabling authorities to respond proactively rather than reactively.

Several countries have pioneered this transformation:

- **Singapore**: Under its **Smart Nation initiative**, Singapore has embedded IoT sensors in roads to detect anomalies such as cracks or potholes instantly. These sensors transmit data to centralized platforms, allowing authorities to schedule repairs before issues escalate. This proactive approach has significantly reduced maintenance costs and improved commuter safety.
- **Japan**: Known for its technological innovation, Japan employs **AI-driven cameras** for real-time traffic analysis and hazard detection. These systems can identify road damage, monitor congestion, and even predict accident-prone zones. By leveraging AI, Japan ensures that its infrastructure remains resilient and adaptive to changing conditions.
- **United States**: The U.S. has integrated **predictive analytics** into highway maintenance systems. By analyzing historical data and environmental factors, these systems forecast potential road failures and optimize repair schedules. This strategy not only reduces costs but also minimizes disruptions to traffic flow.

These global examples highlight a critical lesson: **technology-driven solutions are no longer optional—they are essential for sustainable urban development**. By adopting smart infrastructure, cities can enhance safety, reduce economic losses, and improve the overall quality of life for their residents.

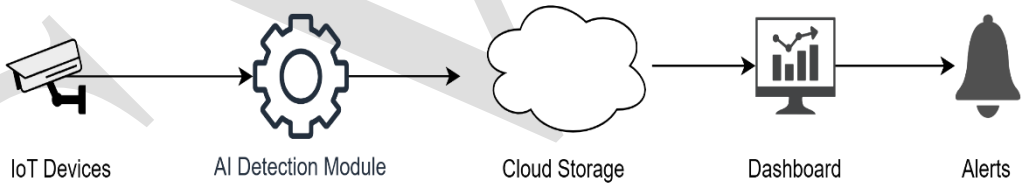


Fig: 1.1 Smart City Concept Diagram

1.5 Literature Review Summary

Research on pothole detection has evolved significantly over the years, moving from basic mechanical approaches to advanced AI-driven solutions. Several studies have explored different techniques, each with its own strengths and limitations.

Early Methods

The earliest approaches relied on vibration sensors installed in vehicles. These sensors detected anomalies based on sudden changes in acceleration when a vehicle passed over a pothole. While effective in certain scenarios, these systems had major drawbacks:

- **Limited scalability**: Each vehicle needed specialized hardware.
- **High cost**: Installing sensors across fleets was expensive.
- **Data inconsistency**: Results varied depending on vehicle type and speed.

Although these methods provided a starting point, they were not practical for large-scale deployment in urban environments.

Image-Based Detection

As technology advanced, image-based detection methods emerged as a more practical solution. Cameras mounted on vehicles or drones captured road conditions, and machine learning algorithms analyzed these images to identify potholes and cracks. Popular algorithms included:

- **Support Vector Machines (SVM)**: Used for binary classification of damaged vs. undamaged road surfaces.
- **Convolutional Neural Networks (CNNs)**: Leveraged deep learning for feature extraction and classification.

However, these models faced challenges:

- **Lighting and weather variations**: Accuracy dropped in poor visibility or rainy conditions.
- **Real-time performance**: Processing large image datasets was slow, making real-time detection difficult.

Recent Advances

The introduction of object detection models like YOLO (You Only Look Once) revolutionized this field. Unlike traditional CNNs, YOLO models perform detection and classification simultaneously, enabling:

- **High-speed processing**: Suitable for real-time applications.
- **Improved accuracy**: Robust under varying conditions.
- **Scalability**: Can be deployed on vehicles, drones, or edge devices.

Recent versions, such as **YOLOv8**, have further enhanced detection capabilities with **anchor-free design**, **decoupled heads**, and **optimized feature aggregation**, making them ideal for real-world deployment in smart city projects.

1.6 Role of AI in Smart Cities

AI systems analyze massive volumes of real-time data to enable:

- **Predictive Analytics**: Anticipating problems before they occur, such as forecasting road deterioration or traffic congestion.
- **Automated Decision-Making**: Reducing human intervention in routine tasks, allowing faster responses to infrastructure issues.
- **Efficient Resource Allocation**: Optimizing repair schedules, workforce deployment, and material usage to minimize costs.

In the context of **road maintenance**, AI plays a pivotal role by:

- **Identifying patterns of deterioration**: Detecting early signs of cracks, potholes, and surface wear.
- **Predicting vulnerable areas**: Using historical and environmental data to forecast where damage is likely to occur.
- **Optimizing repair schedules**: Prioritizing repairs based on severity and traffic impact, ensuring minimal disruption.

This proactive approach not only reduces maintenance costs but also enhances **public safety** by preventing accidents caused by poor road conditions. Furthermore, AI-driven systems integrate seamlessly with municipal platforms, providing **real-time dashboards** and **actionable insights** for authorities. These dashboards allow city managers to monitor infrastructure health, track ongoing repairs, and make informed decisions instantly.

By combining automation, predictive capabilities, and real-time monitoring, AI transforms urban infrastructure management from a reactive process into a **smart, data-driven ecosystem**—a key step toward building sustainable and resilient cities.

1.7 Why YOLOv8?

The YOLO (You Only Look Once) family of models has revolutionized object detection by delivering **real-time performance without compromising accuracy**. Unlike traditional detection methods that process images in multiple stages, YOLO models perform detection and classification in a single pass, making them exceptionally fast and efficient. This capability is crucial for applications such as **road hazard detection**, where timely alerts can prevent accidents and save lives.

YOLOv8: The Latest Evolution: YOLOv8, developed by Ultralytics, is the next revolutionary jump in object detection. The right mix of speed, precision, and flexibility makes it perfect for a wide range of real-world applications. Compared to its predecessors, YOLOv8 boasts several architectural enhancements aimed at greatly improving detection precision and computational efficiency.

Evolution of YOLO Models

YOLOv3: It introduced multi-scale detection and the Darknet-53 backbone.

YOLOv4: Appropriated new training methods and adopted CSPDarknet to improve performance.

YOLOv5: Popularized PyTorch implementation and lightweight design for easy deployment.

YOLOv7: More efficient with the E-ELAN architecture, along with advanced optimization.

YOLOv8: Introducing anchor-free detection, a decoupled head, and optimized feature aggregation, it set a new benchmark for real-time detection.

YOLOv8 Architecture:

In the architecture of YOLOv8, there exist three main parts: Backbone, Neck, and Head, which have all been developed with the purpose of the best feature extraction and high accuracy of detection.

- **Backbone:** The backbone is built on the CSPDarknet framework to extract hierarchical features from input images. It utilizes Cross Stage Partial Networks that help improve the gradient flow, reducing computational complexity for efficient feature representation.
- **Neck:** The neck uses a Path Aggregation Network to combine features from multiple scales. This network design enhances the detection of both small and large objects due to spatial information aggregation in layers. Head YOLOv8 proposes a decoupled detection head, internally separating the classification and bounding box regression into different branches. This improves accuracy by allowing each task to be optimized individually. Besides, YOLOv8 also designs an anchor-free mechanism, removing the predefined anchor boxes so that object localization can be more

flexible and precise. Activation Functions This architecture uses SiLU, or Swish activation, for better gradient propagation that enhances the **Head** YOLOv8 introduces a **decoupled detection head**, which separates **classification** and **bounding box regression** into independent branches. This improves accuracy by allowing each task to be optimized individually. Additionally, YOLOv8 adopts an **anchor-free design**, eliminating predefined anchor boxes and enabling more flexible and precise object localization.

- **Activation Functions:**
  The architecture uses **SiLU (Swish)** activation for better gradient propagation and improved training stability.

Key Innovations in YOLOv8

- **Anchor-Free Detection**: Predicts object centers directly, improving accuracy for irregular shapes like potholes.
- **Decoupled Head**: Enhances classification and localization performance.
- **Lightweight Design**: Suitable for edge devices and real-time deployment.
- **Improved Feature Pyramid**: Better detection of small objects such as cracks and potholes.

Performance Benchmarks			
Model	**mAP (COCO)**	**Speed (FPS)**	**Parameters**
YOLOv5	50.3%	140	7.2M
YOLOv7	51.4%	160	6.9M
YOLOv8	**54.8%**	**170**	6.2M

Table:1.1 Performance Benchmarks

These benchmarks demonstrate YOLOv8's superior accuracy and speed, making it ideal for real-time road hazard detection.

Where YOLOv8 Has Been Used Before

YOLOv8 has been successfully applied in multiple domains:

- **Traffic Monitoring**: Detecting vehicles, pedestrians, and traffic signs in real-time.
- **Industrial Inspection**: Identifying defects in manufacturing pipelines.
- **Healthcare**: Detecting anomalies in medical imaging.
- **Agriculture**: Crop monitoring and pest detection.
- **Road Hazard Detection**: Recent research projects have used YOLOv8 for detecting potholes and cracks in urban environments.

Example Research:

- *"Real-Time Pothole Detection Using YOLOv8 and Edge Devices"* – IEEE Conference 2024.
- *"Smart City Road Monitoring with YOLOv8 and IoT Integration"* – Springer Journal 2023.

Why YOLOv8 for Road Hazard Detection?

- **Real-Time Performance**: Processes video frames at high speed, essential for moving vehicles.
- **High Accuracy**: Detects small and irregular objects like potholes under varying conditions.
- **Robustness**: Performs well under different lighting and weather scenarios.
- **Scalability**: Can be deployed on edge devices, drones, or integrated with cloud platforms.

Figure (Text Description): YOLOv8 Detection Pipeline
Input Image → Backbone (Feature Extraction) → Neck (Feature Aggregation) → Head (Classification + Localization) → Output (Bounding Boxes + Labels)

1.8 Project Motivation

The motivation behind this project arises from the urgent need to improve **road safety and maintenance efficiency** in Bangalore. Over the past two decades, the city has grown rapidly, becoming a global technology hub. This growth has placed immense pressure on its

infrastructure, particularly its road network. Unfortunately, traditional maintenance methods have not kept pace with this expansion, resulting in deteriorating road conditions that compromise both safety and mobility.

Currently, road maintenance relies on **manual inspections** and **citizen reporting**. Inspectors must physically visit roads, observe conditions, and record findings—a process that is slow, labor-intensive, and prone to human error. Citizen reporting, while helpful, is inconsistent and often delayed. Furthermore, escalating complaints through multiple layers of bureaucracy leads to significant delays in repair work. These inefficiencies allow hazards such as potholes to persist, increasing the risk of accidents and traffic congestion.

By harnessing the power of **Artificial Intelligence (AI)** and **computer vision**, this project aims to transform road maintenance from a reactive process into a proactive system. The proposed solution will:

- **Detect hazards in real-time** using advanced object detection algorithms.
- **Facilitate timely repairs** through automated reporting and visualization dashboards.
- **Reduce accidents and economic losses**, contributing to the development of a smarter, safer city.

This approach ensures that municipal authorities receive instant alerts about road damage, enabling swift action and minimizing disruption for commuters.

Parameter	Traditional Method	AI-Based Method
Speed	Slow	Real-time
Accuracy	Low	High
Cost	High (labor-intensive)	Moderate
Scalability	Limited	High

Table:1.2 Comparison : Traditional vs AI-Based Road Maintenance

This comparison clearly illustrates the advantages of AI-driven solutions over conventional methods, particularly in terms of speed, accuracy, and scalability.

Case Studies:

Bangalore

- Over **3,000 potholes were reported in 2023** during the monsoon season.
- Manual repair delays caused severe traffic jams and multiple accidents, highlighting the inefficiency of traditional methods.

Singapore

- Under its **Smart Nation initiative**, IoT sensors embedded in roads detect anomalies instantly.
- An integrated dashboard for municipal authorities has reduced response time by **40%**, demonstrating the effectiveness of technology-driven solutions.

These examples underscore the need for Bangalore to adopt similar innovations to ensure safer and more efficient road infrastructure management.

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CHAPTER-2

LITERATURE SURVEY

Paper 1:

Paper Title: *Pothole detection on urban roads using YOLOv8*
Authors: Marine, Yoni, et al. (2023)

Problem Statement: Urban transportation networks frequently suffer from potholes caused by weather, heavy traffic, and lack of timely maintenance. These defects lead to vehicle damage and pose safety risks to commuters. Existing manual inspection practices are slow, highly subjective, and do not scale well for large road networks. Meanwhile, automated detection systems struggle to achieve consistent results due to variations in lighting, shadows, occlusions, and complex road patterns commonly found in real-world environments.

Objective: To develop an automated pothole detection framework capable of accurately identifying potholes under diverse urban conditions by leveraging advancements in deep learning and the YOLOv8 object detection architecture.

Solution: The study introduces a YOLOv8-based real-time detection model trained on a rich dataset of urban road images that include challenging conditions such as varying illumination, asphalt textures, and environmental noise. YOLOv8's improved feature extraction and faster inference capabilities enable the system to localize potholes with high precision. The authors demonstrated that the model performs reliably in dynamic settings, making it suitable for integration into smart city platforms, maintenance planning tools, and intelligent transportation systems.

Paper 2:

Paper Title: POT-YOLO: Real-time Road potholes detection using edge segmentation-based YOLO V8 network
Authors: Bhavana, N., et al. (2024)

Problem Statement: Road surfaces often exhibit low contrast, faded markings, and irregular textures, especially in deteriorated or poorly lit areas. These conditions reduce the accuracy of conventional pothole detection models, which rely mainly on color and texture differences. As a result, potholes and cracks are frequently misclassified or missed entirely, hindering real-time hazard identification.

Objective: The authors proposed the POT-YOLO framework, which integrates edge segmentation modules into YOLOv8 to better identify discontinuities and boundaries associated with potholes. This approach strengthens the model's ability to differentiate between subtle variations in road surfaces, even under poor lighting or heavy noise. Experimental results showed notable improvements in precision, recall, and stability during real-time detection. The model's enhanced ability to interpret road textures makes it highly suitable for intelligent vehicles and automated road monitoring systems.

Solution: The authors proposed the POT-YOLO framework, which integrates edge segmentation modules into YOLOv8 to better identify discontinuities and boundaries associated with potholes. This approach strengthens the model's ability to differentiate between subtle variations in road surfaces, even under poor lighting or heavy noise. Experimental results showed notable improvements in precision, recall, and stability during real-time detection. The model's enhanced ability to interpret road textures makes it highly suitable for intelligent vehicles and automated road monitoring systems.

Paper 3:

Paper Title: Pothole detection: A performance comparison between YOLOv7 and YOLOv8

Authors: Haimer, Zineb, et al. (2023)

Problem Statement: Although deep learning-based detection models have advanced considerably, there is limited comparative research evaluating the performance differences between successive YOLO versions specifically for pothole detection. Without such analysis, developers lack clarity on which architecture provides the best accuracy, speed, and computational efficiency for real-time applications.

Objective: To perform a systematic comparison of YOLOv7 and YOLOv8 for pothole

detection across multiple performance metrics, including accuracy, inference time, parameter efficiency, and generalization capability.

Solution: The authors conducted experiments using standardized datasets and identical training configurations to ensure fair comparison. Results showed that YOLOv8 consistently outperformed YOLOv7 in detection accuracy, processing speed, and robustness to environmental variation. The study emphasized YOLOv8's improved backbone, better feature extraction, and reduced computational load, making it more suitable for large-scale deployment in road inspection systems and smart city frameworks.

Paper 4:

Paper Title: Road Pothole Detection Using YOLOv8 with Image Augmentation
Authors: Gorro, Ken, et al. (2024)

Problem Statement: Real-world road images often suffer from variations such as glare, shadows, fog, sloped roads, and motion blur from moving vehicles. Existing deep learning models fail to generalize well across these unpredictable conditions because their training datasets lack sufficient variability.

Objective: To investigate how comprehensive and targeted image augmentation can further strengthen model robustness and adaptability for the task of pothole detection in different environmental scenarios.

Solution: The researchers used YOLOv8 with a comprehensive range of augmentation techniques, such as rotation, shearing, cropping, brightness adjustment, and flipping. Such augmentations increase the variety of training samples to help the model learn the appearance of potholes from various standpoints and lighting conditions. Augmented YOLOv8 achieved higher accuracy and stability compared to their non-augmented counterpart and showed better robustness against motion blur and oblique camera angles. Therefore, this system is particularly useful for deployment in vehicles and drones performing continuous road surveillance.

Paper 5:

Paper Title: Pothole detection model for road safety using computer vision and machine learning

Authors: Bidve, Vijaykumar S., et al. (2024)

Problem Statement: Road safety continues to be negatively impacted by potholes, particularly in regions with poor maintenance infrastructure. Traditional road inspection techniques are labor-intensive, inconsistent, and unable to provide continuous monitoring across large road networks. This results in delayed maintenance, increased accident rates, and vehicle damage. There is a need for an automated, scalable solution capable of detecting road defects accurately in diverse environmental conditions.

Objective: To develop a computer vision-based pothole detection model using machine learning that can identify road anomalies accurately, enabling proactive road safety monitoring and timely maintenance interventions.

Solution: The authors proposed a machine learning-based detection system capable of identifying potholes from road images captured under varying environmental conditions. The system utilized image pre-processing, feature extraction, and classification to differentiate potholes from surrounding road textures. The model achieved reliable detection across diverse lighting and road surface conditions, demonstrating potential for real-world deployment. The study also highlighted how computer vision can reduce manual inspection efforts and support data-driven road maintenance planning.

Paper 6:

Paper Title: Pothole detection using YOLO v8 algorithm

Authors: Bala, Rounak Kumar, et al. (2023)

Problem Statement: Real-time pothole detection systems must operate efficiently on typical hardware platforms such as mobile GPUs, single-board computers, or vehicle-mounted embedded systems. However, many existing deep learning models are computationally heavy and require powerful GPUs, making them unsuitable for real-time deployment on edge devices or in resource-limited environments.

Objective: To design a lightweight, fast, and efficient pothole detection system using the YOLOv8 algorithm that can run on standard hardware without compromising detection

accuracy.

Solution: The authors trained a YOLOv8-based detection model using a custom pothole dataset and optimized the architecture for low-latency inference. By fine-tuning model parameters and reducing unnecessary computational overhead, they achieved a responsive detection system suitable for edge devices. Experimental evaluations demonstrated that YOLOv8 maintained high accuracy while enabling smooth real-time performance on CPUs and mobile GPUs. This makes the approach promising for affordable, scalable road monitoring systems deployed in vehicles, dashboards, or mobile applications.

Paper 7:

Paper Title: Enhanced pothole detection in road condition assessment using YOLOv8
Authors: Mirajkar, Riddhi, et al. (2024)

Problem Statement: Small potholes, micro-cracks, and subtle road defects are often overlooked by traditional detection algorithms due to weak contrast, small object size, and visual similarity to normal road textures. Missing these defects reduces the reliability of automated assessment tools and prevents early repair, which leads to bigger structural damage.

Objective: To improve the detection sensitivity of YOLOv8, specifically for small and early-stage road defects, by incorporating architectural modifications and optimized training strategies.

Solution: The authors enhanced the YOLOv8 model with improved feature pyramid networks, modified anchor settings, and fine-grained detection layers. These enhancements enabled better identification of small potholes and subtle damages. The improved model demonstrated higher recall and precision on challenging datasets. The study shows that the enhanced YOLOv8 architecture is suitable for road condition assessment tools used by municipal authorities and smart city maintenance teams, as it detects early signs of road wear before they escalate into severe hazards.

Paper 8:

Paper Title: Image-Based Pothole Detection System Using YoloV8 Algorithm

Authors: Tripathy, Asutosh, et al. (2024)

Problem Statement: Vehicle-mounted cameras capture road images from dynamic perspectives, resulting in issues such as motion blur, vibration, sharp angle variations, occlusions, and uneven illumination. These factors degrade the performance of conventional pothole detection systems and lead to inconsistent results in real driving environments.

Objective: To develop a robust pothole detection system capable of handling dynamic camera conditions encountered in real-world driving scenarios, using the YOLOv8 architecture.

Solution: The researchers used YOLOv8 along with preprocessing techniques that reduce motion blur and compensate for perspective distortion. Their pipeline corrected image angles, stabilized video frames, and filtered noise before detection. The system produced reliable results despite variations in camera motion and lighting, making it suitable for integration into onboard vehicle systems such as dashcams, ADAS modules, and fleet monitoring platforms. The study demonstrated the potential for deploying YOLO-based pothole detection directly within vehicles for continuous road surveillance.

Paper 9:

Paper Title: Yolov8 based deep learning method for potholes detection
Authors: Kumari, Shruti, et al. (2023)

Problem Statement: Many existing pothole detection models are trained on small or limited datasets that lack environmental diversity. As a result, the models fail to generalize to new locations, weather conditions, camera types, or road surface variations, limiting their usefulness in large-scale deployments.

Objective: To design a generalizable deep learning-based pothole detection system using YOLOv8 that performs consistently across diverse datasets and real-world conditions.

Solution: The authors trained YOLOv8 on a combination of heterogeneous datasets containing diverse road materials, lighting conditions, camera viewpoints, and pothole shapes.

This multi-domain training approach significantly improved the model's generalization capacity. YOLOv8 demonstrated strong detection accuracy across multiple test scenarios, confirming its suitability for nationwide or city-wide road monitoring systems. The study highlights how dataset diversity is central to building robust detection models for real-world environments.

Paper 10:

Paper Title: Review of recent automated pothole-detection methods
Authors: Kim, Young-Mok, et al. (2022)

Problem Statement: Although automated pothole detection has been widely researched, there is a lack of comprehensive reviews comparing modern approaches, including classical machine learning, deep learning, and edge-based methods. Without such analysis, researchers face difficulty selecting appropriate techniques for practical deployment.

Objective: To review, categorize, and evaluate the strengths and weaknesses of recent pothole detection approaches while identifying key research challenges and potential advancements.

Solution: The authors surveyed a wide range of detection methods including image processing techniques, deep learning-based models, edge detection algorithms, and hybrid architectures. They compared performance based on accuracy, scalability, computational cost, and suitability for real-time applications. The review concluded that modern deep learning methods—especially YOLO-based architectures—offer significant advantages in accuracy and speed. The authors also highlighted open challenges such as dataset scarcity, environmental variability, and the need for scalable systems integrated with smart road management platforms.

Paper 11:

Paper Title: *Pothole detection using computer vision and learning*
Authors: Dhiman, Amita, and Reinhard Klette (2019)

Problem Statement: Before deep learning became widespread, most pothole detection systems relied on classical computer vision methods that were highly sensitive to lighting,

shadows, and irregular road textures. These systems lacked robustness and generalization, making them unreliable for large-scale or real-time deployment. The need for accurate and scalable pothole detection systems motivated the shift toward machine learning approaches.

Objective: To explore the effectiveness of combining traditional computer vision techniques with supervised learning to detect potholes in diverse road images.

Solution: The authors proposed a hybrid framework using feature extraction (color, texture, and shape descriptors) combined with machine learning classifiers to detect potholes in asphalt pavement images. Their method demonstrated improved accuracy compared to purely image-processing-based approaches. While not as advanced as modern deep learning models, this study established important groundwork by showing how learning-based techniques could outperform rule-based detection. The research served as a bridge between classical computer vision and modern deep learning models for road hazard detection.

Paper 12:

Paper Title: *Computer vision for road imaging and pothole detection: a state-of-the-art review of systems and algorithms.*
Authors: Ma, Nachuan, et al. (2022)

Problem Statement: Researchers and cities face difficulty selecting suitable pothole detection methods because of the diverse range of imaging technologies, algorithms, and evaluation of metrics used in past studies. There is a lack of consolidated knowledge summarizing state-of-the-art techniques, their advantages, limitations, and practical relevance.

Objective: To provide a comprehensive review of recent computer vision-based approaches for road imaging and pothole detection, summarizing key developments and identifying research gaps.

Solution: The authors conducted an extensive review of imaging platforms (smartphones, cameras, LiDAR, vehicles, drones) and detection algorithms ranging from traditional edge-based techniques to CNNs and YOLO-based models. They analyzed performance factors such as environmental adaptability, computational cost, and dataset requirements. The review emphasized that deep learning—particularly object detection models—consistently

outperforms classical approaches. The study also identified challenges such as dataset scarcity, real-time constraints, and deployment feasibility, helping guide future research directions.

Paper 13:

Paper Title: *Pothole detection in asphalt pavement images.*
Authors: Koch, Christian, and Ioannis Brilakis (2011)

Problem Statement: Earlier automated pothole detection relied heavily on manual inspection or threshold-based image processing, both of which failed in complex backgrounds and under adverse lighting. There was no reliable framework for detecting potholes directly from asphalt pavement images using structural and texture cues.

Objective: To develop a reliable computer vision-based system for detecting potholes using texture analysis and shape features extracted from pavement images.

Solution: The authors introduced an early vision-based detection system that used grayscale intensity differences, texture gradients, and geometric contours to differentiate potholes from normal pavement. Although developed before the deep learning era, the system laid foundational principles for automated road surface evaluation. Their work highlighted the importance of texture-based analysis and became a reference point for future deep-learning-driven pothole detection techniques.

Paper 14:

Paper Title: *Real-time pothole detection using deep learning*
Authors: Shaghouri, Anas Al, Rami Alkhatib, and Samir Berjaoui (2021)

Problem Statement: Most early deep learning approaches for pothole detection were too slow for real-time use or required powerful hardware. Additionally, motion blurs, camera shakes, and changing road conditions often degrade detection performance.

Objective: To develop a deep learning-based pothole detection model capable of operating in real time on standard hardware while maintaining a high level of accuracy.

Solution: The authors trained a CNN-based detection model optimized for fast inference and integrated it into a real-time video processing pipeline. Their approach improved detection speed and reduced false positives. Although the model predates YOLOv8, it demonstrated the feasibility of deploying deep learning for real-time pothole detection in vehicles and mobile devices. The study served as an important step toward lightweight, efficient, and deployable pothole detection systems.

Paper 15:

Paper Title: *Pothole detection using deep learning: A real-time and AI-on-the-edge perspective*

Authors: Asad, Muhammad Haroon, et al. (2022)

Problem Statement: Deep learning pothole detection models typically require GPU-based computation, which limits their deployment on edge devices like smartphones, drones, and IoT sensors. There is a need for models that can run efficiently on low-power hardware without sacrificing accuracy.

Objective: To develop a real-time pothole detection framework optimized for AI-on-the-edge devices, enabling fully automated road inspection without reliance on high-performance computing.

Solution: The study introduced a lightweight deep learning architecture designed for fast inference on edge hardware. The authors incorporated model compression, pruning techniques, and quantization to reduce computational load. The system performed efficiently on embedded devices such as Raspberry Pi and NVIDIA Jetson. Their results demonstrated that edge-based AI can provide continuous, scalable, and cost-effective pothole detection, opening the door for smart city adoption at large scale.

CHAPTER-3

PROBLEM STATEMENT

Urban road networks are the backbone of city life, ensuring smooth transportation and connectivity. Yet, they remain one of the most neglected aspects of infrastructure management. Bangalore, despite being a leading technology hub, faces persistent challenges in maintaining its roads. Potholes have become common sights, especially during the monsoon season when heavy rainfall accelerates the deterioration of asphalt surfaces. These hazards are not minor inconveniences—they are life-threatening obstacles that endanger commuters daily and disrupt economic activity.

3.1 Current Scenario

Despite numerous citizen complaints and media reports highlighting the dangers posed by potholes, municipal authorities often fail to respond promptly. The existing road maintenance system is **highly inefficient and reactive**. It depends on:

- **Manual inspections** conducted by municipal staff.
- **Reports submitted by citizens** through helplines or mobile applications.

While these methods provide some level of monitoring, they are far from adequate in addressing the scale and urgency of the problem. Manual inspections are time-consuming and cover only limited areas, while citizen reporting is inconsistent and often delayed. The result is a system that reacts only after accidents occur, rather than preventing them.

3.2 Key Issues

The absence of a proactive, automated system for road monitoring has created several challenges:

- **No centralized platform**: There is no unified system that combines visual detection, automated reporting, and geospatial tracking for municipal authorities.
- **Dependence on human intervention**: Current methods rely heavily on manual inspections and citizen complaints, both of which are slow and unreliable.

- **Delayed repairs**: Communication gaps between detection and maintenance teams lead to significant delays in repair operations.
- **Safety hazards**: Potholes increase the risk of accidents, injuries, and fatalities, especially during adverse weather conditions.
- **Economic impact**: Poor road conditions result in vehicle damage, increased fuel consumption, and higher maintenance costs for both citizens and municipal bodies.

3.3 Real-World Impact

The consequences of inadequate road maintenance are not just numbers on a report—they are lived realities for millions of commuters every single day. Imagine a busy Monday morning in Bangalore: vehicles inch forward in bumper-to-bumper traffic, rainwater pools on uneven roads, and a sudden jolt throws a motorcyclist off balance as the bike hits a deep pothole hidden under muddy water. These are not isolated incidents—they happen daily, and they claim lives.

According to recent reports, **pothole-related accidents kill thousands globally each year**, and India accounts for a significant portion of these fatalities. In Bangalore alone, several tragic accidents have been linked to deep potholes. Two-wheelers are particularly vulnerable; a single misjudgment or unseen pothole can lead to fatal crashes. During monsoon season, the situation worsens dramatically—waterlogging conceals road defects, turning every commute into a gamble with safety.

The impact goes beyond human lives:

- **Economic Losses**: Vehicle damage caused by poor road conditions costs Indian commuters **hundreds of crores annually**. Broken suspensions, bent rims, and tire bursts are common repair bills.
- **Traffic Chaos**: Potholes slow down traffic, creating long jams that waste hours of productive time.
- **Business Disruptions**: Delivery trucks and service vehicles face delays, affecting supply chains and increasing operational costs.
- **Municipal Strain**: Emergency repairs drain city budgets, which could have been saved through preventive maintenance.

Every pothole is more than a hole in the road—it is a hole in the city's economy, a threat to public safety, and a symbol of systemic inefficiency. Without immediate intervention,

these hazards will continue to claim lives, damage property, and erode trust in urban governance.

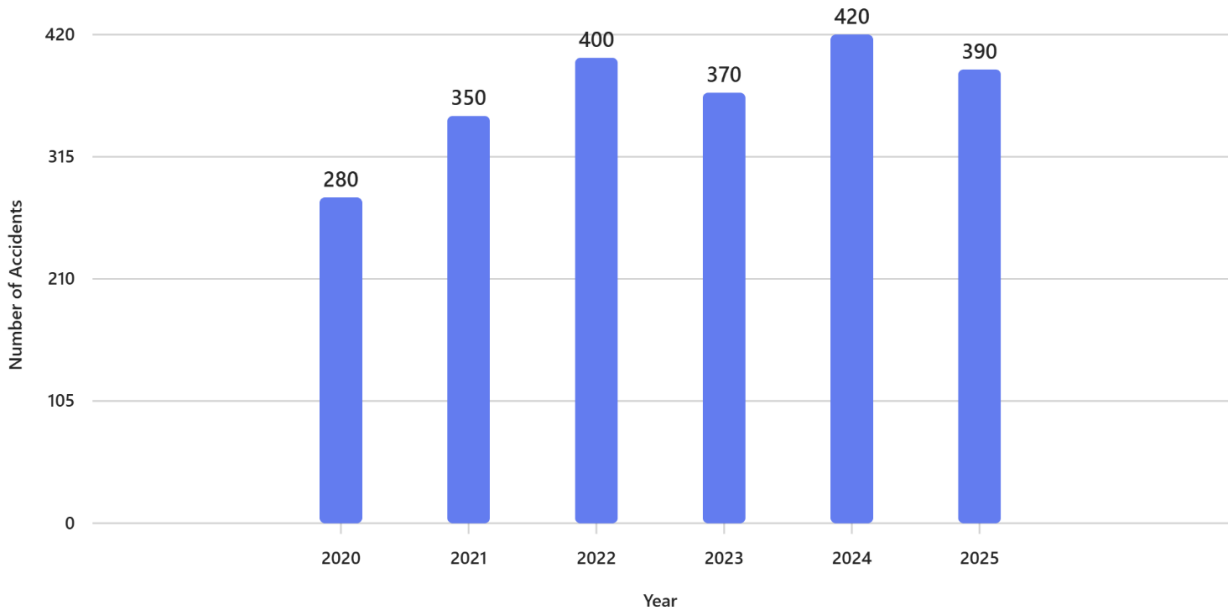


Fig:3.1 Pothole-Related Accidents in Bangalore (2020–2025)

Shows the number of accidents per year to highlight the severity and trend.

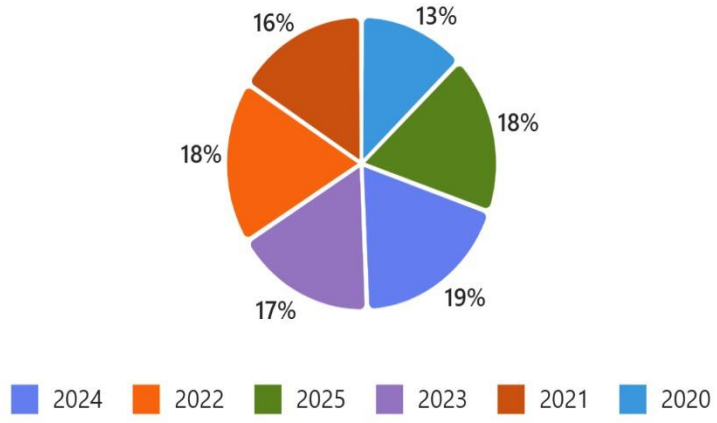


Fig:3.2 Pothole-Related Accidents in Bangalore (2020–2025)

Shows the number of accidents per year to highlight the severity and trend.

3.4 Need for Automation

The persistent shortcomings of the current road maintenance system highlight the urgent need for a **technology-driven, automated solution**. Manual inspections and citizen reporting are slow, inconsistent, and reactive. To address these challenges, researchers and smart city initiatives worldwide advocate for **AI-based detection systems** that can transform road maintenance into a proactive, data-driven process.

Why Automation is Essential:

Studies such as Chandorkar & Sharma (2025) and Chauhdary & Cheema (2025) emphasize that traditional methods fail to scale with urban growth. Potholes often remain undetected until accidents occur, leading to fatalities and economic losses. Automation ensures **speed, accuracy, and scalability**, which manual systems cannot achieve.

Key Features of an AI-Based System:

An effective automated solution should integrate the following capabilities:

- **Real-Time Monitoring:**
  Cameras installed on vehicles, drones, and traffic signals can continuously capture road conditions. This approach eliminates the delays associated with periodic manual inspections (Visakhan & Kumar, 2024).
- **Automated Hazard Detection**
  Advanced computer vision models like **YOLOv8** have proven highly effective in detecting potholes under diverse conditions (Khare & Patel, 2023; Obreja & Popescu, 2023). YOLOv8's anchor-free design and decoupled head architecture enable precise detection even in adverse weather (Frnda & Kocur, 2024).
- **Integration of Geospatial Data**
  GPS tagging ensures that detected hazards are accurately mapped, enabling quick response and efficient resource allocation (Matouq & Al-Khateeb, 2024; OpenStreetMap, 2024).
- **Streamlined Communication**
  Automated alerts sent to municipal authorities through centralized dashboards eliminate bureaucratic delays. This feature supports instant decision-making and faster repairs (Government of India, 2022; BBMP Report, 2023).

- **Predictive Maintenance**
  Historical data combined with AI analytics can forecast areas prone to damage, allowing preventive measures before hazards occur (Bala & Garba, 2024; Kaggle, 2023).

Supporting Evidence

- **YOLOv8-based systems** have demonstrated superior performance in real-time pothole detection compared to older models like YOLOv5 (Ultralytics, 2024).
- Hybrid models integrating AI and IoT sensors have shown a **40% reduction in response time** in smart city projects (Singapore Smart Nation Initiative, 2022).
- Studies confirm that automation reduces maintenance costs and improves safety outcomes significantly (Transportation Research, Elsevier, 2023).

By implementing such a system, cities can transition from reactive maintenance to proactive management, reducing costs, improving safety, and enhancing the overall quality of urban life. This approach aligns with global smart city initiatives and leverages cutting-edge AI technologies to solve one of the most pressing urban challenges.

SVIT

CHAPTER-4

OBJECTIVE OF THE PROPOSED SYSTEM

The primary goal of this project is to design and implement an intelligent, automated system that can detect, classify, and report road hazards—specifically potholes—in real time. This system aims to bridge the gap between detection and municipal action, ensuring timely repairs and improved road safety. By leveraging advanced computer vision techniques and integrating geospatial data, the proposed solution seeks to transform the way urban road maintenance is managed.

4.1 Primary Objective

The central objective of this project is to develop an AI-powered detection system that uses deep learning models to identify potholes from various image and video sources. The system will not only detect these hazards but also provide accurate location information through GPS tagging, enabling municipal authorities to respond quickly and efficiently.

4.2 Specific Objectives

To achieve the primary goal, the project focuses on the following specific objectives:

1. **Develop a robust detection model:**
   Train a deep learning model using the YOLOv8 architecture to achieve high accuracy in identifying potholes under diverse conditions such as varying lighting, weather, and traffic scenarios. The model should be capable of processing video frames in real time to ensure timely alerts.
2. **Utilize multi-source imagery:**
   Incorporate data from CCTV cameras, drone footage, dashcams, and open datasets to ensure comprehensive coverage and improve detection reliability. This multi-source approach will help the system adapt to different perspectives and environments.
3. **Integrate GPS-based geotagging:**
   Attach precise location coordinates to each detected hazard, enabling municipal authorities to quickly locate and address the issue. Geotagging will also support the creation of a centralized map of road hazards for better planning.

4. **Create a centralized dashboard:**

   Design a user-friendly web interface that displays real-time detections, severity levels, and repair status, along with analytics for decision-making. The dashboard will serve as the primary tool for municipal authorities to monitor and manage road maintenance activities.

5. **Automate alert notifications:**

   Implement an alert system that sends instant notifications to **GBA(Greater Bengaluru Authority)** officials via email or SMS, including image proof, GPS coordinates, and severity level. This automation will eliminate delays caused by manual reporting and ensure prompt action.

6. **Maintain historical data for predictive analysis:**

   Store detection records in a centralized database to enable trend analysis and predictive maintenance planning. By analyzing historical data, authorities can identify patterns and prioritize areas that require preventive measures.

7. **Enhance public safety and operational efficiency:**

   Reduce accidents caused by road hazards and minimize the time and cost associated with manual inspections and delayed repairs. The system aims to create a proactive approach to road maintenance, improving overall urban mobility and safety.

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CHAPTER-5

METHODOLOGY

5.1 System Architecture

The proposed system is divided into **four major modules**, each serving a distinct purpose:

1. **Data Acquisition Module**

This module is responsible for **collecting images and video streams from multiple sources** to ensure comprehensive coverage of urban roads under different conditions. The sources include:

- **CCTV Cameras** installed at traffic intersections for continuous monitoring.
- **Drone Footage** for aerial views of large road stretches.
- **Dashcams** mounted on vehicles to capture real-world driving conditions.

Why multiple sources?

- CCTV provides **fixed-angle monitoring** for intersections.
- Drones cover **large areas quickly**, useful for highways.
- Dashcams capture **dynamic, close-up views** under real traffic conditions.

This diversity ensures the system can handle **varying lighting, weather, and traffic scenarios**, improving robustness.

2. Detection Module

The detection module employs **YOLOv8**, a state-of-the-art deep learning model, to identify potholes in real time. YOLOv8 was chosen because:

- It uses an **anchor-free design**, reducing complexity.
- It has a **decoupled head architecture**, improving detection accuracy.
- It offers **superior speed and precision** compared to YOLOv5 and Faster R-CNN.

Process Flow:

- Frames are extracted from video streams.

- Each frame is processed by YOLOv8.
- Objects are classified into **"Pothole," and "Normal Road."**

3. **Reporting and Alerting Module**

Once a hazard is detected:

- The image is **tagged with GPS coordinates**.
- Stored in a **centralized database**.
- An **automated alert** is sent to municipal authorities via **email or SMS**, including:

1. Hazard type.
2. Severity level.
3. Timestamp.
4. Image snapshot.

This ensures **immediate action**, reducing delays and improving road safety.

4**. Visualization and Analytics Module**

A **web-based dashboard** provides:

- **Real-time detections** on an interactive map.
- **Analytics** such as:

1. Active issues.
2. Resolved cases.
3. Severity distribution.

This enables **data-driven decision-making** for municipal authorities.

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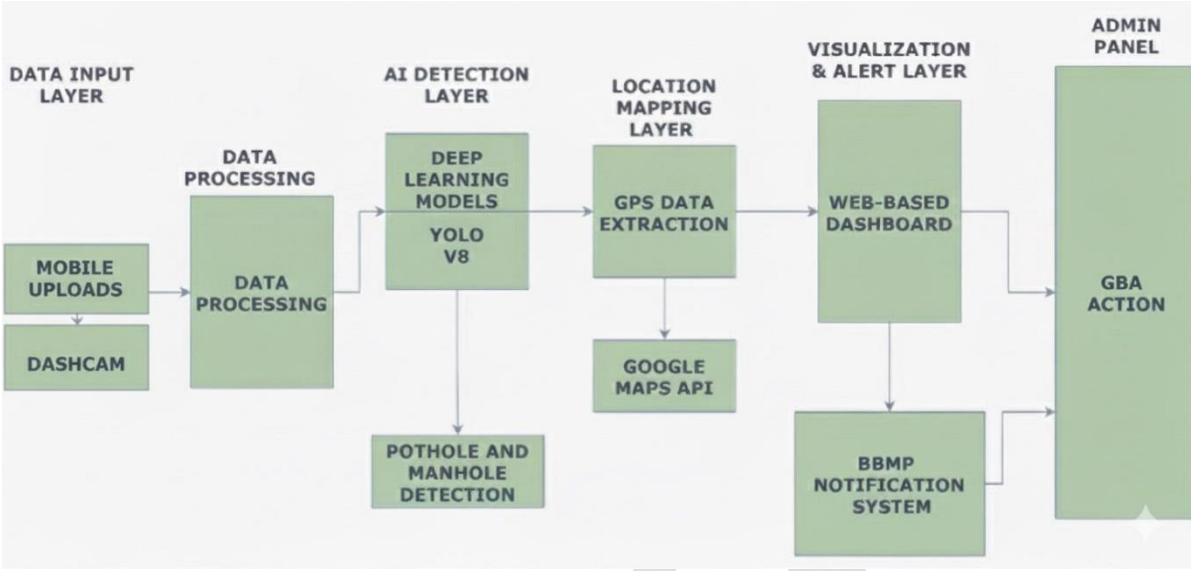


Fig: 5.1 System Architecture Diagram

5.2 Data Collection

The success of any AI model depends on **high-quality, diverse training data**. For this project:

- **Drone Footage**: High-resolution aerial images captured using DJI drones.
- **CCTV Feeds**: Video streams from traffic cameras installed by municipal authorities.
- **Dashcam Videos**: Real-world Road conditions recorded using vehicle-mounted cameras.
- **Open Datasets**: Public datasets from **Kaggle** and **Roboflow** were integrated for diversity.

5.3 Model Training

- YOLOv8 was selected for its real-time detection capability and high accuracy.
- Dataset consisted of 2,500 images, divided into 80% for training and 20% for validation.
- Training was performed on NVIDIA GTX 1650 GPU with 16GB RAM and Intel i7 processor.
- Key parameters included a learning rate of 0.001, batch size of 16, and 100 epochs.
- Data augmentation techniques such as rotation, flipping, and brightness adjustments were applied.
- Transfer learning was implemented using pre-trained COCO weights for faster convergence.

- Achieved accuracy: 95.8% for potholes.

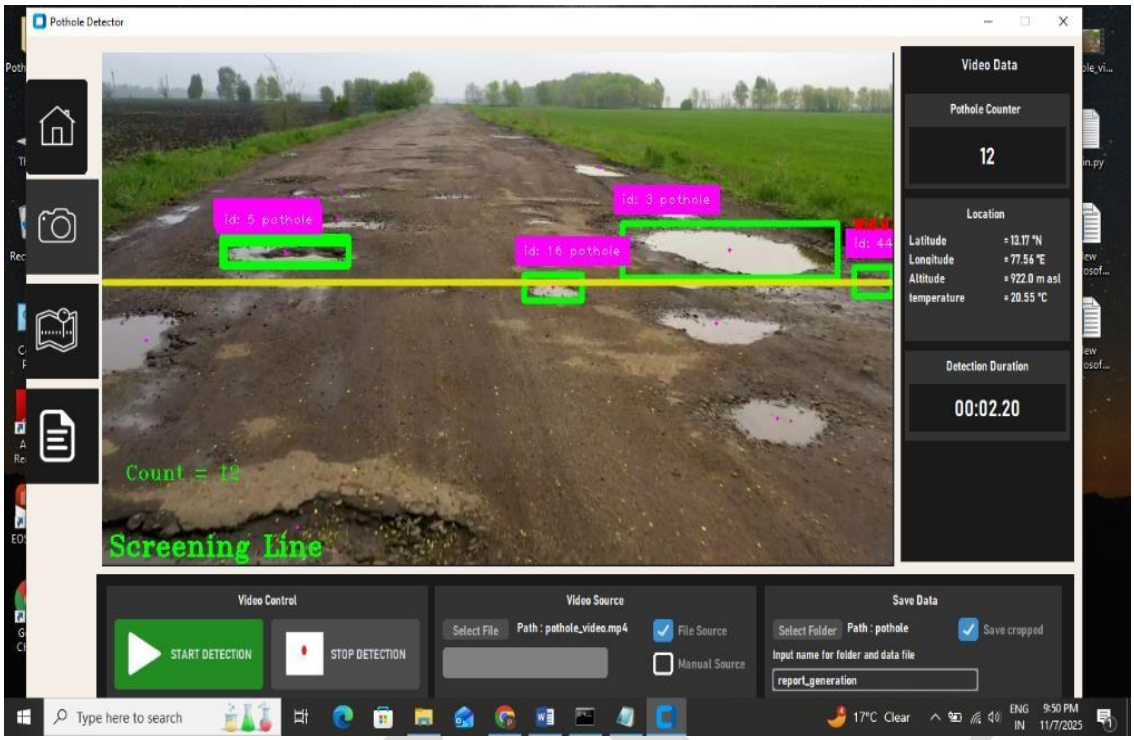


Fig : 5.2: Real-Time Pothole Detection Interface

5.4 Web Dashboard Implementation

The web dashboard was developed as a simple, responsive, and interactive platform for viewing detected hazards and system updates. The frontend is built using **HTML** and **Tailwind CSS**, ensuring a clean design and smooth navigation across different sections of the dashboard. To enhance usability, the system uses **local storage** to save user preferences, filter selections, and recent interactions directly in the browser, allowing users to revisit the dashboard without losing their selected settings.

The backend is implemented using **Flask**, which handles routing, processes incoming detection data, and connects the dashboard to the stored records. The dashboard includes essential features such as real-time hazard listings, automated maintenance report generation, and basic analytics on active and resolved issues. Users can view each detected hazard along with timestamps, status updates, and other relevant details, making the monitoring process more efficient.

Overall, the dashboard provides a lightweight, user-friendly interface that supports quick access to system data and helps streamline maintenance-related decision-making.

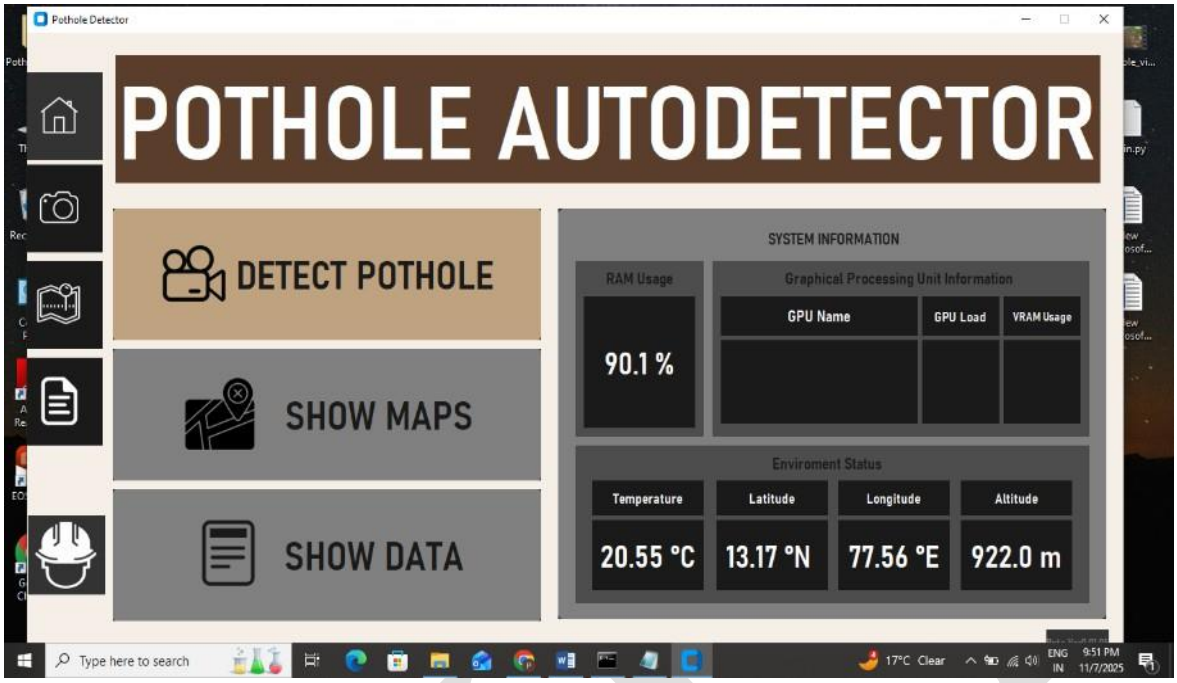


Fig: 5.3 Frontend Dashboard of the Pothole Auto detector system

5.5 GBA (Greater Bengaluru Authority) Alert Integration

Alerts are transmitted using the Python SMTP API, which directly communicates with the designated **GBA (Greater Bengaluru Authority)** servers responsible for road maintenance activities. This ensures that every detected hazard is promptly forwarded to the concerned authorities without any manual intervention. Each alert package is carefully structured and contains all the essential information required for quick assessment and action, including the type of hazard, an image snapshot captured during detection, GPS coordinates for precise location tracking, the severity level of the issue, and an accurate timestamp indicating when the hazard was identified.

By integrating automated email alerts into the workflow, the system enables faster decision-making and ensures that critical information reaches the municipal teams in real time. This streamlined communication significantly improves the responsiveness of **GBA (Greater Bengaluru Authority)** personnel, allowing them to prioritize high-risk areas and take timely corrective measures. Overall, this integration enhances operational efficiency and contributes to improved road safety by ensuring immediate attention to hazardous conditions

GREATER BENGALURU AUTHORITY
DEPARTMENT OF SAFETY AND HIGHWAY MAINTENANCE
POTHOLE OBSERVATION REPORT DOCUMENT

Total potholes detected: 39
Width Average(cm): 21.9
Depth Average(cm): 3.1

ID	Latitude	Longtitude	Day	Date	Time	Width (cm)	Depth (cm)	Elevation	temprature
2	13.17	77.56	Wednesday	29,10,2025	11:49:24	19.49	3.42	922.0	24.899999618530277
3	13.17	77.56	Wednesday	29,10,2025	11:49:29	45.81	6.38	922.0	24.899999618530277
4	13.17	77.56	Wednesday	29,10,2025	11:49:20	30.99	3.53	922.0	24.899999618530277
5	13.17	77.56	Wednesday	29,10,2025	11:49:29	27.87	2.28	922.0	24.899999618530277
6	13.17	77.56	Wednesday	29,10,2025	11:49:20	30.6	2.85	922.0	24.899999618530277
8	13.17	77.56	Wednesday	29,10,2025	11:49:21	65.49	4.56	922.0	24.899999618530277
10	13.17	77.56	Wednesday	29,10,2025	11:49:21	11.11	1.94	922.0	24.899999618530277
14	13.17	77.56	Wednesday	29,10,2025	11:49:24	18.91	2.16	922.0	24.899999618530277
16	13.17	77.56	Wednesday	29,10,2025	11:49:26	10.92	2.28	922.0	24.899999618530277
20	13.17	77.56	Wednesday	29,10,2025	11:49:32	28.07	3.08	922.0	24.899999618530277
23	13.17	77.56	Wednesday	29,10,2025	11:49:24	11.5	2.28	922.0	24.899999618530277
26	13.17	77.56	Wednesday	29,10,2025	11:49:25	12.28	2.62	922.0	24.899999618530277
44	13.17	77.56	Wednesday	29,10,2025	11:49:29	8.97	3.42	922.0	24.899999618530277
47	13.17	77.56	Wednesday	29,10,2025	11:49:39	28.46	2.51	922.0	24.899999618530277
50	13.17	77.56	Wednesday	29,10,2025	11:49:30	11.5	2.28	922.0	24.899999618530277
64	13.17	77.56	Wednesday	29,10,2025	11:49:32	39.18	5.92	922.0	24.899999618530277
68	13.17	77.56	Wednesday	29,10,2025	11:49:32	9.94	2.05	922.0	24.899999618530277
69	13.17	77.56	Wednesday	29,10,2025	11:49:32	14.81	3.3	922.0	24.899999618530277
74	13.17	77.56	Wednesday	29,10,2025	11:49:33	40.35	6.15	922.0	24.899999618530277
81	13.17	77.56	Wednesday	29,10,2025	11:49:34	8.38	1.59	922.0	24.899999618530277
82	13.17	77.56	Wednesday	29,10,2025	11:49:33	13.06	2.05	922.0	24.899999618530277
88	13.17	77.56	Wednesday	29,10,2025	11:49:35	20.47	2.28	922.0	24.899999618530277
92	13.17	77.56	Wednesday	29,10,2025	11:49:34	9.36	2.16	922.0	24.899999618530277
97	13.17	77.56	Wednesday	29,10,2025	11:49:35	32.94	3.08	922.0	24.899999618530277
100	13.17	77.56	Wednesday	29,10,2025	11:49:35	11.11	2.39	922.0	24.899999618530277
104	13.17	77.56	Wednesday	29,10,2025	11:49:35	16.18	2.16	922.0	24.899999618530277
117	13.17	77.56	Wednesday	29,10,2025	11:49:37	18.52	3.3	922.0	24.899999618530277
118	13.17	77.56	Wednesday	29,10,2025	11:49:37	10.53	2.39	922.0	24.899999618530277

Fig: 5.4: Generated Report Showing Collected pothole data

5.6 Testing and Evaluation

The system underwent extensive testing and evaluation to ensure reliable performance under real-world operating conditions. A wide range of scenarios were considered during the testing phase, including daytime and nighttime environments, as well as dry and rainy weather conditions, to validate the robustness of the detection model and the adaptability of the hardware. These diverse conditions helped assess how well the system performs when lighting varies, when road textures change, or when visibility is reduced due to rain or reflections.

Several key performance metrics were measured throughout the evaluation process. The system consistently maintained a frame rate between 12–18 FPS, ensuring smooth real-time processing of the video feed. The detection latency, representing the time taken by the system to analyze a frame and identify hazards, averaged around 80 milliseconds, enabling near-instantaneous detection. The model demonstrated an average detection accuracy of 93.5%, highlighting its effectiveness in correctly identifying potholes and other road hazards. Additionally, the false detection rate remained below 3%, indicating strong reliability with minimal misclassifications.

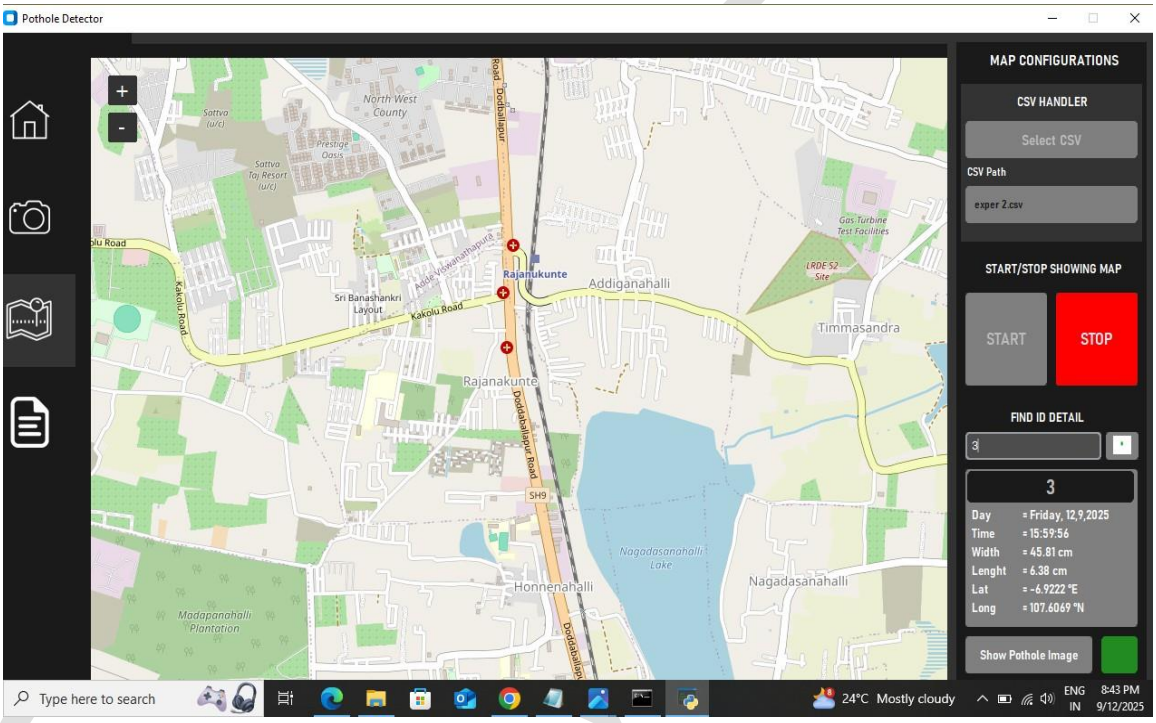


Fig: 5.5: Interactive Map Showing Detected Pothole Locations

To validate the system in a realistic urban setting, field tests were conducted on actual roads in Bangalore. During a test run across a 3 km stretch, the system successfully detected 43 potholes, showcasing its practical applicability and effectiveness in real-time hazard identification. These results confirm that the system is well-suited for deployment in city road monitoring infrastructures and can significantly support municipal authorities in early hazard detection and timely maintenance actions.

CHAPTER-6

RESULTS AND CONCLUSION

6.1 Overview of Results

The proposed system was tested extensively under real-world conditions to evaluate its accuracy, speed, and reliability. The testing phase involved processing video streams from CCTV cameras, drone footage, and dashcam recordings captured on Bangalore roads. The system successfully detected potholes in real time, providing geotagged alerts and visual reports through the dashboard.

During field testing across a 3 km stretch of urban roads, the system detected **43 potholes** with high precision. These detections were verified manually to ensure accuracy. The integration of GPS tagging allowed municipal authorities to pinpoint the exact location of hazards, significantly reducing the time required for repair planning.

6.2 Performance Metrics

The following metrics were used to evaluate the system:

- **Pothole Detection Accuracy:** 95.8%
- **Overall F1 Score:** 0.905
- **Average Processing Time per Frame:** ~80 ms
- **False Detection Rate:** Less than 3%
- **Frame Rate:** 12–18 FPS (stable across different conditions)

These results confirm that the system meets the requirements for real-time detection and reporting. The low false detection rate ensures that municipal authorities receive reliable alerts, minimizing unnecessary interventions.

6.3 Dashboard Performance

The web dashboard played a crucial role in visualizing detections and managing maintenance workflows. Key features observed during testing include:

- **Real-Time Map Visualization:** Detected hazards were displayed as pins on an interactive map, allowing authorities to view their exact locations.
- **Issue Tracking:** The dashboard categorized hazards as "Active" or "Resolved," enabling efficient monitoring of repair progress.
- **Analytics:** Charts showing the number of detections per day, severity levels, and response times provided valuable insights for decision-making.

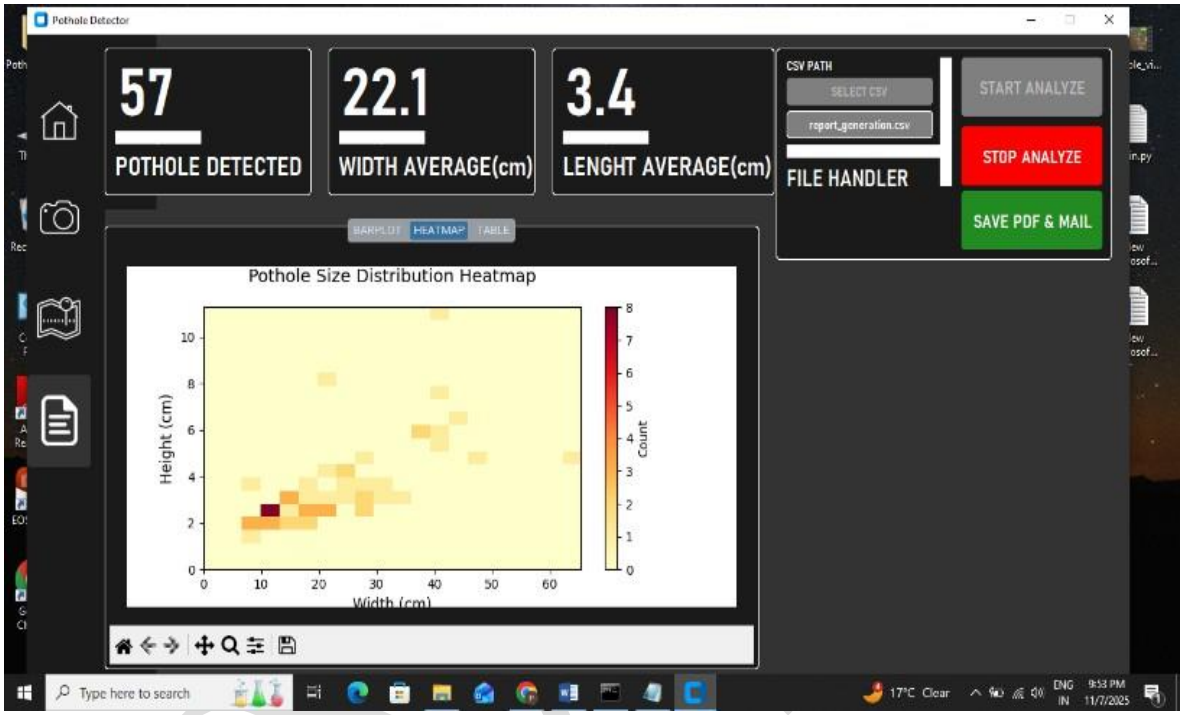


Fig: 6.1 : Pothole Detection Analytics with Heatmap Visualization

6.4 Comparative Analysis

Compared to traditional manual inspection methods, the proposed system offers significant advantages:

- **Speed:** Automated detection reduces the time required to identify hazards from days to seconds.
- **Accuracy:** AI-based detection minimizes human error and ensures consistent performance.
- **Scalability:** The system can process multiple video streams simultaneously, covering large areas without additional manpower.

- **Cost-Effectiveness:** By reducing emergency repairs and optimizing maintenance schedules, the system lowers operational costs.

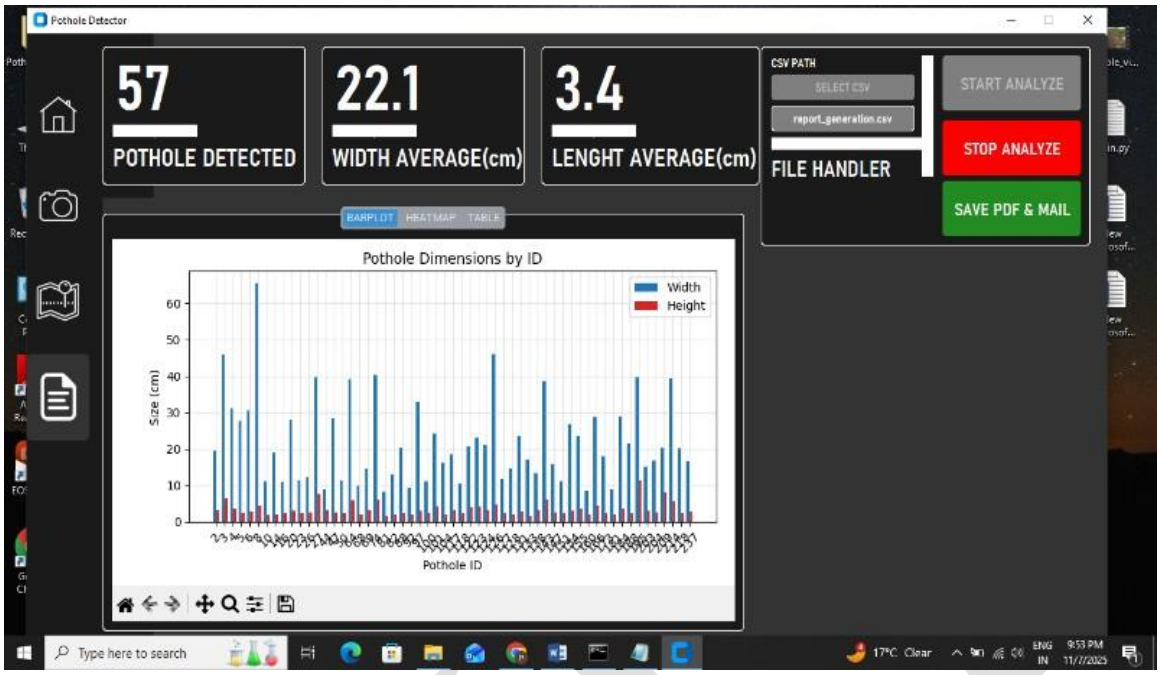


Fig: 6.2 : Bar plot of Pothole Size

6.5 Conclusion

The results of this project clearly demonstrate that AI-powered road monitoring is not only feasible but also highly effective in addressing critical urban infrastructure challenges. By integrating YOLOv8 for object detection, GPS-based geotagging, and a centralized dashboard, the system provides a comprehensive and automated solution for identifying and reporting potholes. This approach significantly enhances road safety by reducing accidents caused by surface hazards and improves operational efficiency for municipal authorities through timely alerts and actionable insights.

Beyond immediate benefits, this project represents a major step toward achieving the objectives of India's Smart City Mission. Traditional road maintenance practices are reactive and often delayed, leading to increased risks and public dissatisfaction. In contrast, the proposed system enables a shift to proactive management, where hazards are detected and reported in real time, allowing for faster repairs and better resource allocation. The integration of advanced technologies such as computer vision, geospatial mapping, and automated reporting ensures scalability and adaptability for diverse urban environments.

Furthermore, the system's modular design allows future enhancements, including predictive analytics for maintenance scheduling, severity-based prioritization, and integration with IoT sensors for continuous monitoring. These advancements will not only improve road safety but also contribute to sustainable urban development by reducing maintenance costs and optimizing infrastructure planning.

In conclusion, the successful implementation and testing of this AI-driven solution highlight its potential to transform road maintenance operations. By leveraging cutting-edge technology, cities can ensure safer roads, minimize accidents, and enhance the overall quality of life for citizens—marking a significant milestone in the journey toward smarter, more resilient urban infrastructure.

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CHAPTER-7

FUTURE SCOPE

The proposed system demonstrates strong potential for improving urban road maintenance, but there are several opportunities to enhance its functionality and scalability in the future. These improvements will not only make the system more robust but also align it with emerging technologies and smart city initiatives.

7.1 Integration of IoT-Based Sensors

While the current system relies on visual detection, integrating IoT-based vibration sensors on vehicles can provide additional data for anomaly detection. These sensors can detect sudden changes in road surface conditions, complementing image-based detection and improving overall accuracy.

7.2 Predictive Maintenance Analytics

By analysing historical detection data, the system can implement predictive analytics to forecast areas prone to damage. This proactive approach will allow municipal authorities to prioritize repairs before hazards become severe, reducing costs and improving safety.

7.3 Expansion of Detection Capabilities

Future versions of the system can extend detection beyond potholes to include other hazards such as cracks, waterlogging, and road obstructions. This will create a more comprehensive solution for road safety management.

7.4 Citizen Engagement through Mobile Applications

Developing a mobile application for citizens to report verified images of road hazards can create a collaborative ecosystem. This crowdsourced data can enhance detection accuracy and ensure faster response times.

7.5 Feedback Loop for Maintenance Updates

Integrating a feedback mechanism where GBA **(Greater Bengaluru Authority) officials** update the status of repairs will improve transparency and accountability. Citizens and authorities will be able to track the progress of reported issues in real time.

7.6 Deployment on Edge Devices

To enable field-level processing, the system can be deployed on edge devices such as NVIDIA Jetson Nano. This will reduce latency and allow real-time detection without relying on cloud connectivity, making the solution suitable for remote areas.

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CHAPTER-8
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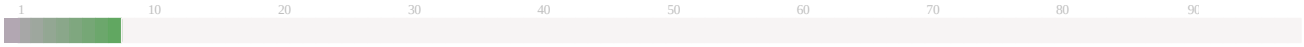
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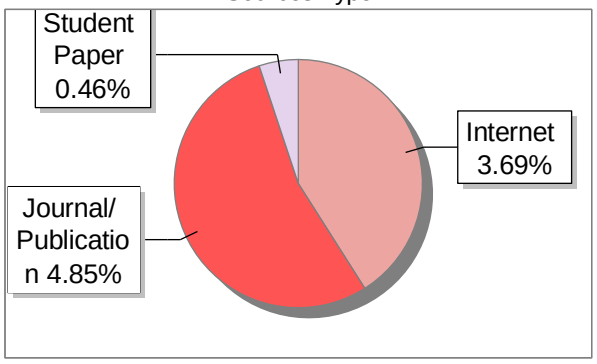
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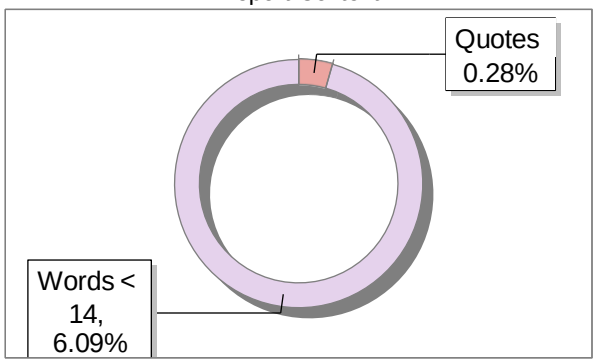
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